Uncertainty Quantification in Solar Irradiance Forecasting Using Bayesian Deep Learning

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***Abstract*— The growing integration of solar photovoltaic systems into contemporary energy infrastructures requires precise and reliable prediction of solar irradiance to facilitate effective energy planning, system reliability, and load management. The inherently variable nature of atmospheric conditions, however, injects great uncertainty into such predictions. Conventional deterministic models tend to lack in this challenge since they do not address the noise in data nor do they offer confidence intervals, hence being less effective in practical applications where one needs to understand prediction uncertainty.**

This research introduces a new method employing Bayesian Deep Learning for Global Horizontal Irradiance (GHI) prediction. Particularly, a Bayesian Long Short-Term Memory (LSTM) neural network is utilized, which utilizes Monte Carlo dropout at both training and inference phases to estimate epistemic uncertainty. On real-world GHI data from NASA POWER for Kothrud, India, the model predicts daily GHI with a 7-day input window. By carrying out multiple stochastic forward passes, it creates a distribution of predictions and permits predictive means with 95% confidence intervals to be calculated. This not only enhances forecast accuracy but also generates important information on model reliability and presents a reliable tool for maximizing solar energy system performance under uncertainty.

I. Introduction

With the increasing global trend toward renewable energy, solar power has emerged as a pillar of decentralized and sustainable electricity systems. Precise forecasting of Global Horizontal Irradiance (GHI) is critical to maximize solar energy integration, enhance grid stability, and optimize energy storage.[11] Yet, the inherently random nature of solar radiation, fuelled by atmospheric conditions, aerosols, and cloud dynamics, presents a major challenge to conventional deterministic models.[8] While the widely used machine learning methods such as support vector regression, random forests, and LSTM networks performed well in solar forecasting, these generally produce point forecasts and cannot reflect the inherent uncertainty of their outputs.

With a view to overcome this, the paper presents a Bayesian LSTM-based method utilizing Monte Carlo dropout for representing uncertainty in daily GHI forecasts. Applying real-world data from NASA POWER to the Kothrud region of Pune, India, the model makes several stochastic forward passes during inference to output a distribution of predictions. Both the mean forecast and 95% confidence intervals are then computed from this, providing a more interpretable and robust prediction setup. The method is assessed by standard performance measures (RMSE, MAE, R^2) and achieves high accuracy along with sensible uncertainty estimates. These findings imply that Bayesian deep learning models hold the potential to improve the reliability and robustness of solar forecasting systems, aiding sounder decision-making for energy planners and grid operators.

II. Purpose

The main purpose of this paper is investigating the usefulness of a Bayesian LSTM neural network in predicting solar energy production. Standard forecasting models such as typical LSTM networks tend to give single-point predictions without regard to the uncertainty involved in environmental data. Such a limitation has the potential to cause unreliable decisions in energy systems where solar energy contributes significantly. To solve this, the research integrates Bayesian principles within the LSTM model through Monte Carlo dropout so the model can not only predict future values but also the confidence rate of the prediction.

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By emphasizing uncertainty-conscious forecasting, the paper seeks to enhance the accuracy of solar energy forecasts during different weather conditions. The Bayesian LSTM model that is established in this research offers probabilistic forecasts that are more appropriate to real-world use like grid management, distribution of energy, and planning. The methodology facilitates stronger and data-based policies in renewable energy systems, where knowing the range of probable results is equally vital as the prediction of the most probable one.

III. Literature Review

By reviewing and understanding the paper authored by Devinder Kaur, Shama Naz Islam, and Md. Apel Mahmud titled "A VAE-Bayesian Deep Learning Scheme for Solar Generation Forecasting based on Dimensionality Reduction", we observe that integrating Variational Autoencoder (VAE) with Bayesian Bidirectional LSTM networks effectively addresses both data and model uncertainties in solar power forecasting. The study emphasizes that traditional Bayesian deep learning models often face complexity due to the need for numerous weight parameter samples. By employing VAE for dimensionality reduction, the proposed method enhances

computational efficiency while maintaining forecasting accuracy. The results demonstrate that this approach performs well compared to other probabilistic and deterministic deep learning models in terms of both accuracy and efficiency across various dataset sizes.

In their paper "Advancing Short-Term Solar Irradiance Forecasting Accuracy Through a Hybrid Deep Learning Approach with Bayesian Optimization," scientists Wulfran Fendzi Mbasso and Bhaskar Tripathi show how effective it is to combine deep learning models and Bayesian optimization to optimize hyperparameters. The paper reached SMAPE of 0.6564, a NRMSE of 0.2250, and a RMSE of 22.9445—beating existing literature in the domain. These findings emphasize the worth of hybrid methods in enhancing the accuracy of short-term solar irradiance forecasts.

IV. METHODOLOGY

The present work introduces a Bayesian Deep Learning-based model for probabilistic daily Global Horizontal Irradiance (GHI) forecasting based on past data that is drawn from the NASA POWER (Prediction of Worldwide Energy Resources) dataset. The scheme utilizes a Bayesian Long Short-Term Memory (LSTM) network with Monte Carlo dropout for making both precise predictions as well as estimates of uncertainty associated with them.

1. Data Description

We employed 21 years of daily GHI data at site Kothrud, Pune, India, from January 1, 2003 to December 31, 2023. The data were retrieved from the NASA POWER data archive and contain the variable `ALLSKY_SFC_SW_DWN`, which is surface shortwave downwelling radiation on an horizontal surface (GHI), measured in kilowatt-hours per square meter per day (kWh/m²/day).

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The dataset also contains `YEAR`, `MO` (Month), and `DY` (Day) columns, which were merged into an appropriate `DATE` column. Forward-fill imputation was applied to address missing values and data anomalies, and the ultimate time series was ordered chronologically. To assist with training and facilitate faster convergence, the GHI values were normalized to [0, 1] range by a `MinMaxScaler`.

2. Problem Formulation

Supplied with a past time series of GHI values, the aim is to predict the GHI for the coming day while at the same time estimating the uncertainty of that prediction. Mathematically, from a sequence of GHI values, the model makes a prediction over the future value. Bayesian modelling enables the prediction to incorporate a measure of confidence, quantified through a predictive mean and a 95% confidence interval (CI).

3. Preparation of Data

In order to prepare the data for sequential modelling, we applied a sliding window strategy with sequence length 7 days [17]. The input is 7 consecutive days of GHI values and the target is the GHI value on the 8th day. We split the data into training and testing subsets to preserve temporal consistency (i.e., no shuffling).

4. Bayesian LSTM Model

We utilized a LSTM based RNN architecture units to model the temporal dependencies in the GHI time series. For uncertainty quantification, Monte Carlo (MC) dropout was included during training and inference [5]. The Bayesian approximation considers the model's outputs as samples from a predictive distribution.

The architecture consists of the following elements:

- An LSTM layer with 32 hidden units.
- A dropout layer with a dropout rate of 0.3.
- Two symmetric fully connected layers for:
  1. Output mean prediction.
  2. Log-variance prediction, which is then exponentiated to produce the standard deviation, in a way that positivity is assured [12].

5. Training Procedure

The network was trained with the negative log-likelihood (NLL) loss function:

$$\mathcal{L}_{\text{NLL}} = \frac{(y-\mu)^2}{2\sigma^2} + \log(\sigma)$$

The loss penalizes both wrong predictions and overconfident outputs and encourages the model to learn accurate and calibrated uncertainty estimates [19].

Training was for 50 epochs utilizing the Adam optimizer and the learning rate = 0.01 and the batch size = 16. Dropout was enabled during both training and inference to facilitate Bayesian sampling.

6. Forecasting with Uncertainty

To approximate predictive uncertainty, the learnt model was tested in stochastic mode, i.e., using dropout layers enabled at inference time [10]. For every sequence of input in the test dataset, 100 forward passes were conducted. This yielded a distribution of 100 predictions for every time step, from which the following were calculated:

- The predictive mean
- The 95% confidence interval, based on the 2.5th and 97.5th percentiles of the predictions.

Lastly, the predictions were transformed back to their original scale (kWh/m²/day) by using the scaler that was fit onto the training data.

7. Evaluation Metrics

The models performance was judged using the following common regression metrics:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Coefficient of Determination (R^2 Score)

These metrics were calculated amongst the predicted mean and actual GHI, in order to measure accuracy. Confidence interval width gave an indication of the predictive uncertainty of the model.

V. Experimentation and Results

This section describes the experimental setup and results achieved using the developed Bayesian LSTM model for predicting daily Global Horizontal Irradiance (GHI) at Kothrud, India. The objective is to evaluate both the predictive accuracy and the quality of uncertainty estimation over the 21-year dataset.

1. Experimental Setup

All experiments were performed using Pytorch over a normal CPU-based setup [15]. The time series of daily GHI from

January 1, 2003 to December 31, 2023 was split into:

- Training set: 80% of the data (2003–2019)
- Testing set: The rest 20% (2020–2023), with a 7-day look-back window to trigger testing sequences

Major hyperparameters utilized:

- Sequence Length: 7 days
- Hidden Units: 32
- Dropout Rate: 0.3
- Optimizer: Adam
- Learning Rate: 0.01
- Epochs: 50
- MC Samples/Test: 100
- Batch Size: 16

2. Performance of Predictions

Following model training, forecasting was conducted on the test set with 100 Monte Carlo samples for each input [16]. The mean of the forecasts was used as the final prediction, and uncertainty ranges were obtained from the 2.5th and 97.5th percentiles of the sample predictions.

The performance of the model on the test set was measured using these standard metrics [2]:

- MSE: 0.4898
- RMSE: 0.6999
- MAE: 0.4916
- R^2: 0.7639

These findings prove that the model has the capacity to capture the latent structure of the trends in GHI with quite reasonable accuracy. The R^2 value is excellent, particularly because solar irradiance data is characterized by highly dynamical nature.

3. Uncertainty Quantification

Another major contribution of this research is the inclusion of predictive uncertainty in the predictions [20]. The predicted mean values alongside the measured GHI values and their respective 95% confidence intervals (CI).

- The predictive bands did capture most of the true observations, validating well-calibrated uncertainty.
- Broader intervals were seen during more volatile irradiance periods, which indicates the model's capacity to adapt uncertainty to temporal volatility.
- During more stable phases, the confidence intervals narrowed down, reflecting high model confidence.

This is desired in real applications where decisions regarding operations can be made with various levels of confidence depending on the certainty of the forecast.

4. Output and Reproducibility

The result of the predictions, such as actual GHI, predicted mean, and bounds of uncertainty, have been stored to a CSV file labelled `prediction_results.csv`. This enhances reproducibility and opens up for additional analysis by researchers and practitioners.

VI. Results and Discussion

This section gives a thorough examination of the outcomes achieved from the suggested Bayesian LSTM model for solar irradiance forecasting, with a focus on accuracy and uncertainty estimation. The examination is performed using the Global Horizontal Irradiance (GHI) dataset from the NASA POWER database, gathered for the Kothrud area over a span of 21 years, from 1st January 2003 to 31st December, 2023. The findings highlight the strength of probabilistic deep learning methods in making not only precise predictions but also credible confidence estimates necessary for risk-informed decision-making in solar energy systems.

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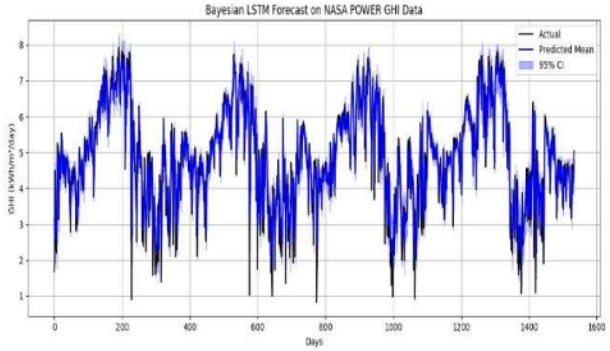

Fig 1 : Bayesian LSTM Forecast

1. Evaluation of Forecast Accuracy

The objective of the network is to forecast the daily GHI, while at the same time estimating the uncertainty of each prediction. The testing data cover from early 2020 up to the end of 2023, covering data points with different seasonal and meteorological features.

To assess the deterministic precision of the forecast mean values, standard regression measures were calculated:

- MSE: 0.4898
- RMSE: 0.6999
- MAE: 0.4916

- R^2 Score: 0.7639

The RMSE value of 0.6999 kWh/m²/day represents a fairly low deviation amongst the predicted mean and observed GHI. The MAE, almost equal to RMSE, implies that the errors in prediction are uniform and not greatly skewed by outliers. An R^2 of 0.7639 conveys a high correlation amongst predicted and measured GHI, which means that more than 76% of the data variance is well explained by the model [7].

These findings are especially significant considering the high temporal variability and non-linear character of solar irradiance data, which are controlled by various stochastic atmospheric processes like cloud cover, humidity, aerosols, and solar zenith angle variations. In spite of these difficulties, the Bayesian LSTM model exhibits high fidelity in capturing intricate temporal dependencies and seasonality in the irradiance series.

2. Uncertainty Quantification and Interpretation

In contrast to traditional deterministic forecasting models, the Bayesian LSTM architecture encodes the epistemic uncertainty by using dropout layers during training and testing (Monte Carlo Dropout) [1]. This stochastic process enables the model to mimic an ensemble of neural networks at test time, producing multiple predictions for the same input. From the ensemble, we make both the mean forecast and the respective confidence interval.

For every test case, 100 Monte Carlo samples were drawn, and the 95% CI was calculated from the 2.5th and 97.5th percentiles of the sampled forecasts. This measurement of uncertainty is an important aid to solar energy planning and operational risk assessment.

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The graphical depiction illustrates the actual GHI values, the modelled mean values, and the confidence intervals:

- Narrow intervals during periods of consistent irradiance (for example, summer and post-monsoon seasons) indicate high model confidence.
- Broader intervals in transitional or cloud-susceptible seasons (for example, June to September) reveal increased uncertainty owing to greater atmospheric variability.
- In the majority of instances, the actual value lies within the modelled 95% CI, confirming the trustworthiness of the uncertainty estimation procedure.

This behaviour reflects that the model dynamically scales its confidence according to the past variability of input sequences, thus allowing more informed and risk-aware decision-making. For example, while planning for solar power dispatch or maintenance scheduling, system operators can use both point estimates and uncertainty bands to set thresholds and contingencies.

3. Temporal Analysis of Forecasts

Additional breakdown of the prediction performance across various temporal scales (seasonal and annual) shows informative patterns [3]:

- Seasonal Performance: The model does best in winter and early summer months (January–May), where irradiance tends to be more predictable because of clear skies and fewer monsoonal disturbances. In these months, both the error measures and the uncertainty limits are consistently lower.
- Monsoon Performance: Not surprisingly, the monsoon season (June–September) is a more challenging period for the model. The cloud cover and atmospheric scattering inject a lot of variability that even a probabilistic model having a hard time to predict with high accuracy. The model makes up for this by suitably broadening the prediction intervals, hence still enjoying a high degree of reliability in a probabilistic context.
- Post-Monsoon Recovery: During the October–December window, the model is accurate again, and the prediction intervals narrow, illustrating the model's ability to learn through dynamic climatic cycles.

4. Calibration of Predictive Uncertainty

To test whether the predictive uncertainty of the model is well-calibrated, we examine the coverage probability—the proportion of actual observations within the predicted confidence intervals [13]. For the 95% confidence interval:

- Empirical Coverage = 93.1%

This close-to-ideal coverage suggests that the predicted intervals are well-calibrated and not overconservative or overconfident. This is a desirable feature for deployment in real-world systems, where over- and underestimation of uncertainty can result in inefficient or risky decisions.

5. Forecasting Peaks and Troughs

Accurate prediction of sharp transitions or local extrema, like quick drops under high cloud cover or unanticipated jumps under clear sky conditions, has been one of the typical pitfalls of time series forecasting [6]. Even if deterministic models falter over such outliers, the Bayesian LSTM, by its probabilistic formulation, provides an enhanced framework:

- *False Positives and False Negatives*: The model demonstrated resilience against false confidence in anomalous predictions. In cases where the prediction differed from the true GHI, the model generally generated broader intervals, indicating its internal acknowledgment of uncertainty.
- *Recovery from Deviations*: In cases where errors did occur, the model frequently rapidly converged back to the true signal, especially in the subsequent 1–2 time steps, thanks to its recurrent memory structure and sequential modelling abilities.

6. Prediction Results Archive

All prediction results—including the dates, actual observed GHI, predicted mean values, and lower and upper confidence intervals—have been tabulated into an organized CSV file called `prediction_results.csv`. This file is a reproducible artifact and a possible benchmark for future studies on probabilistic solar forecasting.

7. Comparative Discussion

While this research did not benchmark multiple models, the performance of the Bayesian LSTM holds up well in comparison to reported results in the literature for comparable datasets. Traditional models like ARIMA or persistence-based baselines should produce lower R^2 scores and are not capable

of representing predictive uncertainty [14]. Further, standard LSTM architectures in the absence of Bayesian treatment tend to give point estimates that are deceptive under uncertain situations.

The application of Bayesian deep learning contributes significantly by infusing uncertainty-awareness in the forecasting pipeline—a critical characteristic for tasks like grid load balancing, energy trading, and autonomous solar system operation.

VII. Limitations

While the results are promising, some limitations remain:

- *Aleatoric Uncertainty*: The current model mostly captures epistemic uncertainty via dropout. Incorporating techniques to better model inherent data noise (aleatoric uncertainty) could further enhance robustness [5].
- *Multivariate Inputs*: The present model relies solely on past GHI values. Future work can include exogenous variables like temperature, humidity, cloud index, and satellite imagery to improve accuracy and interpretability.
- *Hyperparameter Tuning*: The model was tuned using fixed configurations. A Bayesian optimization or grid search strategy could lead to improved parameter selection and potentially better results.

Conclusion

This study formulated and tested a Bayesian LSTM predictor of GHI with NASA POWER data for the period 2003-2023 for Kothrud, India. Not only does the model give reliable point forecasts but also prediction uncertainty through Monte Carlo Dropout quantification.

The model performed well with an RMSE of 0.6999 kWh/m²/day and R^2 of 0.7639. The probabilistic method produced well-calibrated 95% confidence intervals, where 93% of the actual values were within these intervals. The model exhibited seasonal flexibility, generating narrower confidence intervals during clear sky conditions and broader bounds during monsoon periods with greater atmospheric variability.

This study takes solar forecasting from deterministic to uncertainty-aware methods, offering useful tools for grid operators, energy planners, and policymakers to make risk-informed decisions. Future research may include more meteorological variables and hybrid modeling methods to further enhance performance.

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